SPECIALIZED SCHEDULER & NOTIFICATION SYSTEM

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1.0 Abstract

Balancing physiotherapy sessions with academic responsibilities poses a significant challenge for students, particularly those living with physical disabilities, such as those attending King George VI center. To mitigate this challenge, we propose a Specialized Scheduler and Notification Mobile Application. This application enables seamless scheduling of therapy sessions for students and physiotherapists, ensuring alignment with academic schedules. By allowing students to input their academic timetable and physiotherapists to view students' available time slots, the system facilitates the scheduling of sessions during free periods. Automated reminders for upcoming sessions are sent to both students and physiotherapists, while students can track their session attendance progress. Developed using Java in Android Studio, this application offers personalized notifications, scheduling functionalities, and collaborative tools. It enhances time management, organization, and planning, while also supporting progress tracking, motivation, and communication. Ultimately, the proposed system aims to alleviate the challenges faced by students in balancing physiotherapy and academic commitments, fostering improved well-being and academic success.

Keywords: Scheduling, notifications, sessions, progress tracking, communication, appointment, physiotherapist

2.0. Introduction

Students living with physical disabilities at King George VI center often face challenges in managing their healthcare needs (physiotherapy sessions) alongside their academic responsibilities. Physiotherapy sessions are crucial for their well-being, but scheduling and managing these sessions can be difficult due to conflicting academic schedules, forgetfulness, and a lack of personalized time management tools. The proposed tool is a Specialized Scheduler and Notification Mobile Application that allows students and physiotherapist to schedule therapy sessions without conflicting with the academic schedule

3.0. Problem definition

Students with physical challenges particularly at King George VI center often face significant difficulties in balancing their academic responsibilities with essential physiotherapy sessions. Traditional scheduling systems do not adequately account for the unique needs of these students, leading to several problems where existing scheduling systems fail to integrate academic and physiotherapy timetables, resulting in frequent conflicts. Students are often forced to choose between attending classes and their therapy sessions, compromising their academic performance and health outcomes. There is a lack of effective communication tools within current systems, making it difficult for students to interact with their physiotherapists. Missed Appointments, without automated reminders, students and physiotherapists often forget scheduled sessions.

Addressing these issues, the Specialized Scheduler and Notification System aims to provide a comprehensive solution that integrates academic and physiotherapy schedules, enhances communication, ensures timely reminders and facilitates progress tracking. This system is essential to improve the overall management of physiotherapy sessions, supporting the academic and health needs of students with physical challenges.

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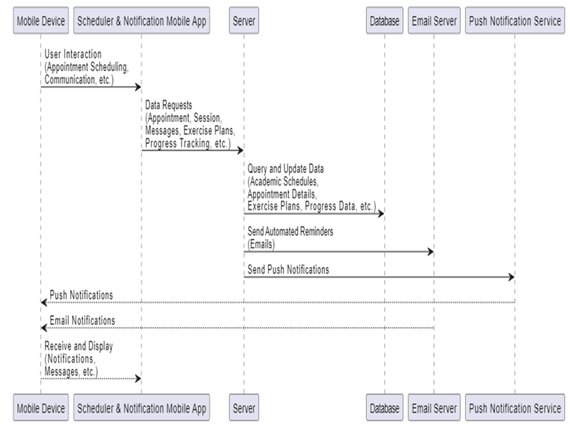
4.0. Overview of the developed application

The developed Specialized Scheduler and Notification System is an innovative solution designed to enhance the coordination and management of physiotherapy sessions for students, particularly those with physical challenges. This system integrates seamlessly with academic schedules and offers a comprehensive suite of features aimed at improving the efficiency and effectiveness of physiotherapy care. Key components of the system include, the system boasts an intuitive interface that allows students to schedule physiotherapy sessions without conflicting with their academic responsibilities. The system provides a communication channel for students to interact directly with their physiotherapists. The system sends automated reminders via email and push notifications. Both students and physiotherapists receive timely alerts about upcoming sessions, helping them stay on schedule. Physiotherapists can create and share tailored exercise plans specific to each student's rehabilitation needs. The developed Specialized Scheduler and Notification System significantly enhances the management and coordination of physiotherapy sessions. By providing a user-friendly, secure, and comprehensive platform, it supports the health and academic success of students, ensuring they receive effective and timely physiotherapy care.

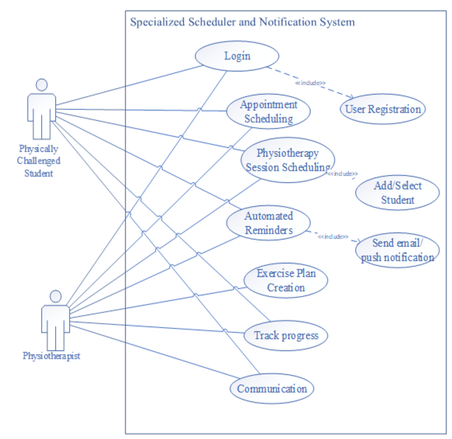
5.0. Objectives

To design a user-friendly interface for appointment scheduling that specifically considers the academic schedules of the student to avoid conflicts with physiotherapy sessions.
To allows the physiotherapists to schedule sessions with their patients/ students
To allows students to easily communicate with their physiotherapist for clarifications, questions, or updates
To allow automated reminders through email, or push notifications to ensure students and physiotherapist are reminded of their upcoming physiotherapy sessions.
To create exercise plans tailored to each student's needs which they can share with their patients
To allows physically challenged students to securely login and access their personalized physiotherapy exercise plans as shared by their physiotherapist
To allow users to track and monitor their physiotherapy attendance progress to enable adjustments to their physiotherapy attendance schedules.

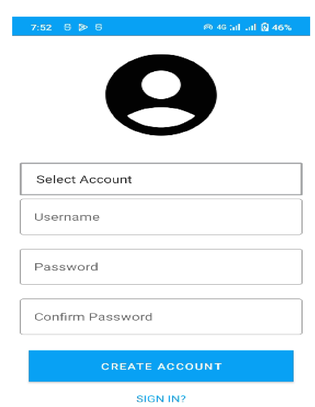
6.0. System Overview

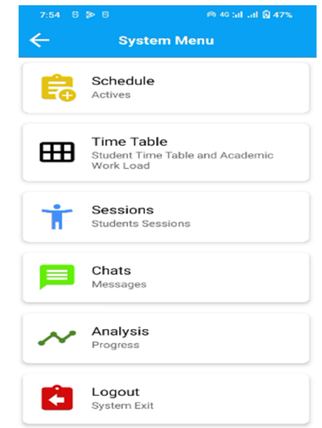


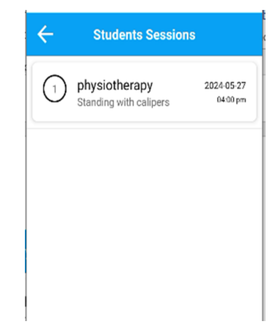
7.0. Use Case Diagram



8.0. Screenshots







Conclusion

In conclusion, the proposed Specialized Scheduler and notification application offers a comprehensive solution for managing physiotherapy sessions, communication, and exercise plans for physically challenged students. By considering the students' academic schedule and workload system minimizes conflicts and optimizes session scheduling. The system's user-friendly interface, automated reminders, and secure communication channels enhance the overall experience for both physiotherapists and students. The personalized exercise plans and progress tracking features enable tailored care and informed treatment adjustments. The proposed system architecture ensures scalability, security, and maintainability, making it a viable solution for educational institutions. By streamlining physiotherapy management, this system has the potential to improve the lives of physically challenged students at King George VI center, enabling them to manage their conditions more effectively and achieve their academic goals. Future development directions include integrating with existing student information systems, expanding to cater to other healthcare professionals, and incorporating artificial intelligence to enhance exercise plan generation and progress

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